Module 6: Operative Vaginal Delivery Versus Second-Stage Cesarean

Companion reading handout for simultaneous listening.
This is Module 6 of Delivery-Room Labor Failures: operative vaginal delivery versus second-stage cesarean.

The previous lecture taught how the uterus and the fetal heart must be read together. This lecture moves to the moment when continued pushing is no longer enough, and the team must choose a route to birth.

That decision is one of the most technically demanding judgments in obstetrics. A low fetal head, an exhausted mother, and a nonreassuring tracing create a powerful desire to act. But action is not one thing. It may mean continued pushing with better mechanics. It may mean manual rotation. It may mean vacuum. It may mean forceps. It may mean cesarean. And each option has a different physiology, a different technical requirement, and a different failure mode.

The purpose of this lecture is not to romanticize operative vaginal delivery. It is also not to surrender every difficult second stage to cesarean. The purpose is to teach the conditions under which an instrument is a precise obstetric tool, and the conditions under which it becomes an unsafe attempt to solve an unsolved mechanical problem.

Here is the central idea.

An operative vaginal instrument should assist a birth whose mechanics are already plausibly deliverable. It should not be used to discover, through force, whether the pelvis, position, station, and fetal size are compatible with birth.

That sentence is worth dwelling on. The vacuum or forceps does not remove the need to understand the passenger, passage, powers, and position. It makes that understanding more important, because the instrument transmits force to the fetal head and maternal tissues. If your diagnosis is correct, the force is guided along the birth canal and delivery occurs. If your diagnosis is wrong, the instrument magnifies the error.

This is why operative vaginal delivery belongs immediately after the lecture on second-stage mechanics.

ACOG frames the issue clearly. Before cesarean delivery for second-stage arrest, clinicians should assess whether operative vaginal delivery is appropriate. That does not mean "try an instrument before cesarean." It means that operative vaginal delivery, when the candidate is appropriate and the operator is skilled, is a legitimate alternative to second-stage cesarean.

The reason is not trivial. Second-stage cesarean is not just ordinary cesarean performed a little later. With the head low in the pelvis, extraction may be more difficult, extensions of the uterine incision are more common, infection risk is higher, and future pregnancies now carry the consequences of a uterine scar. ACOG cites data showing higher endometritis after second-stage cesarean compared with first-stage cesarean, and other data showing increased hysterotomy extensions. The Israeli position paper makes the same practical point: second-stage cesarean has significant maternal and fetal morbidity and implications for future pregnancies.

At the same time, operative vaginal delivery has its own injuries. Vacuum is associated with cephalohematoma, subgaleal hemorrhage, retinal hemorrhage, hyperbilirubinemia, scalp trauma, and rarely intracranial hemorrhage. Forceps are associated with more maternal birth canal trauma and anal sphincter injury, and may cause facial or scalp injury. Shoulder dystocia is more common in instrumental deliveries. The local vacuum protocol gives concrete ranges: cephalohematoma one to twelve percent, subgaleal hemorrhage three to four percent, intracranial hemorrhage six to fourteen per ten thousand vacuum deliveries, retinal hemorrhage seventeen to thirty eight percent, shoulder dystocia around two percent, hyperbilirubinemia five to twenty percent, birth canal lacerations around ten percent, and obstetric anal sphincter injury one to four percent.

So this is not a choice between danger and safety. It is a choice between two forms of risk, guided by whether vaginal birth is technically likely and whether the team can execute the attempt with discipline.

Now we must speak the language of station and position with precision.

Station is measured in centimeters relative to the ischial spines. Zero station means the leading bony part of the head is at the level of the spines. The word bony is crucial. Caput can descend before the skull does. A swollen scalp at the introitus does not prove that the fetal skull is at an outlet station. If the resident confuses caput with station, the entire procedure is built on a false map.

The local protocol defines outlet as scalp visible at the vaginal introitus without separating the labia, with the bony part of the head pressing on the perineum. Low means the leading bony part is at plus two centimeters or lower. Midpelvis means zero or plus one. High means the leading bony part is above the ischial spines.

Gabbe and the ACOG classification use the same clinical logic. Outlet is the head at or on the perineum, scalp visible, with limited rotation. Low is plus two or lower but not on the pelvic floor. Midforceps is above plus two but engaged. High forceps is not included in the classification. The practical lesson is simple: outlet and low procedures are not the same as midpelvic procedures, and a midpelvic OP or OT delivery is a higher-skill, higher-risk event than an outlet OA delivery.

Position is just as important as station. Station tells you how far the head has descended. Position tells you how the head is oriented in the pelvis. A vacuum cup or forceps blade is not placed on an abstract head; it is placed on a head with an occiput, sagittal suture, fontanelles, flexion or deflexion, asynclitism, molding, and caput. If position is uncertain, the direction of traction and the placement of the instrument are uncertain.

This is why Gabbe states that precise position and station must be known before either forceps or vacuum placement. The local protocol allows ultrasound in borderline cases to verify station and position and to estimate chance of success or failure. It notes that an angle of progression lower than one hundred twenty degrees is highly correlated with instrumental delivery failure. Ultrasound is not a ritual requirement for every case, and ACOG notes that adding ultrasound has not necessarily reduced morbidity in trials, even though it improves correct diagnosis of head position. The mature point is this: if the clinical examination leaves meaningful uncertainty, use the tools that clarify the mechanics before applying force.

Now we can understand the prerequisites. They are not a checklist to recite. They are the safety architecture of the procedure.

The cervix must be fully dilated because traction through an incompletely dilated cervix risks cervical trauma and means the route is not open. Membranes must be ruptured because the instrument must contact the fetal head and the mechanics of descent must be direct. The fetus should be vertex for routine vacuum or forceps application; face presentation is a contraindication to vacuum. The head must be engaged, because an unengaged head means the presenting part has not proven entry into the pelvis. The position must be precisely known because placement and traction depend on it. The maternal pelvis must be clinically adequate for the estimated fetal weight because traction cannot safely overcome true disproportion. Analgesia must be adequate because the procedure is painful and because uncontrolled maternal movement increases danger. The bladder must be empty because a full bladder obstructs descent and is vulnerable to injury. Consent must be obtained because the mother is entitled, even under urgency, to understand why the intervention is being proposed, what the alternatives are, and what failure will mean. The operator must be knowledgeable, and, in Gabbe's beautiful formulation, willing to abandon the procedure if necessary. Necessary staff and equipment must be present, because failure cannot be allowed to become delay.

That phrase, willing to abandon, is one of the deepest safety principles in operative obstetrics. The dangerous operator is not only the operator who lacks technical skill. It is the operator who cannot stop.

Indications also need disciplined wording.

Operative vaginal delivery may be considered for prolonged second stage when the station and mechanics are favorable. It may be considered for suspected fetal compromise when shortening the second stage is appropriate and there is no contraindication. It may be considered for maternal benefit: exhaustion, inability to push effectively, or maternal cardiac, neurologic, respiratory, or muscular disease where shortening the second stage matters.

But the indication does not override the prerequisites. A nonreassuring tracing does not make a high head safe for vacuum. Maternal exhaustion does not make uncertain position acceptable. Prolonged second stage does not make suspected macrosomia irrelevant. The more urgent the indication, the more disciplined the candidate selection must be, because a failed attempt consumes precisely the time and reserve that urgency has made precious.

The local contraindications make this concrete.

Absolute contraindications to vacuum include any condition that does not allow vaginal delivery, any non-vertex presentation including face, gestational age younger than thirty two completed weeks, head above the ischial spines, incomplete dilation, fetal predisposition to fractures such as osteogenesis imperfecta, and known fetal bleeding disorders such as alloimmune thrombocytopenia, hemophilia, or von Willebrand disease. A prior cerebral hemorrhage is listed as a relative contraindication in RCOG language.

Relative contraindications locally include head higher than plus two, though midpelvic vacuum can be considered in special cases such as multiparity or second twin by an experienced senior physician; gestational age thirty two to thirty four weeks, where vacuum should generally be avoided and attempted only under special senior circumstances with a low threshold to abandon; suspected macrosomia; significant caput before the procedure; and maternal carriage of HIV, HCV, or HBV.

These details should not live only in a protocol folder. They should live in the resident's hand before the cup touches the head.

Instrument choice is the next layer.

Vacuum and forceps are both operative vaginal delivery. They share the same basic prerequisites. Gabbe explicitly reminds us to treat the vacuum extractor with the same respect as forceps. That is necessary because vacuum can feel deceptively simple. The device is smaller, the application may seem easier, and in many modern units residents receive more vacuum exposure than forceps exposure. But simplicity of appearance does not erase the risk of wrong indication, wrong placement, wrong traction vector, or prolonged attempt.

Forceps and vacuum differ in tradeoffs. Forceps can provide powerful traction and rotation in skilled hands and may be chosen in circumstances where immediate controlled delivery is needed or where the operator's forceps skill is high. Vacuum generally causes less maternal soft tissue trauma than forceps but has more scalp-related neonatal complications and is vulnerable to detachment when placement, traction, or mechanics are poor. The choice should be based on operator experience, fetal position, station, urgency, maternal tissues, fetal condition, and the specific clinical problem.

This is where humility matters. A procedure that is safe in the hands of a skilled operator may be unsafe in the hands of someone who has only theoretical competence. ACOG emphasizes clinician training, hospital setting, available resources, patient preferences, and candidacy. The local protocol is even more operational: vacuum delivery is performed by an attending or an authorized resident under supervision, and every vacuum-assisted delivery requires the supervising physician immediately beside the physician performing the procedure unless the operator is authorized to perform it independently.

Now let us describe vacuum technique only to the level needed for this course's purpose

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The cup should be attached during the pause between contractions. It should be placed about two centimeters anterior to the posterior fontanelle, centered on the sagittal suture, with no maternal tissue between the cup and the scalp. That point is the flexion point. Correct placement promotes flexion and descent. Incorrect placement promotes deflexion, pop-offs, scalp trauma, and failure.

Traction is performed during a contraction and with maternal pushing. The direction of traction follows the axis of the birth canal according to station. The protocol advises continuous pulling, not rocking movements of the head. It prohibits fundal pressure. Suction pressure is not released between contractions until delivery or abandonment. Episiotomy is selective; there are no data supporting routine episiotomy, though it may be considered in specific circumstances.

The instrument must declare progress.

That is the most important procedural principle. With vacuum traction, the head should descend. Gabbe says descent should occur with each pull, and if no descent occurs after three pulls, the operative attempt should be stopped. The local protocol says that if there is no progress after the first two tractions, cup placement and direction of traction must be reassessed. If there is no head progression by an experienced operator, abandon. The duration of the traction procedure is limited locally to fifteen minutes, with extension only by clinical judgment and senior approval. If the cup detaches three times, the procedure must be stopped. After two detachments, the operator should be replaced by a more senior physician if available; a less experienced operator is replaced after one pop-off.

A pop-off is not just an annoying device event. It is information. It may tell you that cup placement is wrong, traction direction is wrong, suction is inadequate, the head is too deflexed, the station is misread, the pelvis is not favorable, or the procedure is being asked to do too much. The correct response is not stubbornness. It is reassessment and, when criteria are met, abandonment.

This is also why preparation for cesarean must exist before the attempt. If failure will cause a delay in cesarean, that fact changes the decision to attempt vacuum in the first place. The local protocol specifically lists situations in which double setup in the operating room should be considered: first delivery at plus two station, OP position, midpelvis, maternal exhaustion, history of failed vacuum, concern that failure would delay cesarean, and suspected macrosomia with estimated fetal weight over four thousand grams.

The double setup concept teaches the resident an ethical point. You do not begin an attempt and then discover that the rescue route was not ready. The rescue route is part of the indication.

Failed vacuum must not become a prolonged second attempt at vaginal birth. The local protocol states that the vacuum failure rate ranges from two point nine to six point five percent, and that failure is associated with a higher rate of fetal complications. In the event of instrumental delivery failure, delivery must proceed immediately by cesarean section. Gabbe adds another crucial warning: sequential use of vacuum and forceps increases adverse maternal and neonatal outcomes more than the sum of the separate risks. In other words, the second instrument is not a harmless continuation. It compounds trauma and delay.

There is a narrow local exception: if the head is at the vaginal opening, and failure is attributed to technical inability to generate effective pressure, one forceps attempt for lift-up only may be performed by a senior physician. The narrowness of that exception is the teaching. It is not permission for sequential instruments in the ordinary failed vacuum. It is a technical exception for a senior operator in an outlet situation.

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Imagine a nulliparous patient after a long second stage. The head is at plus two, but the position is OP, there is significant caput, estimated fetal weight is over four thousand grams, the tracing is category two, and the operator is junior. This is not simply "head is low, try vacuum." Every detail increases the risk of failure: first delivery at plus two, OP, caput, possible macrosomia, and limited operator experience. The mature decision may be double setup, senior examination, ultrasound if position or station is uncertain, and a very explicit choice between a highly controlled attempt and cesarean.

Now imagine a multiparous patient with a clear OA position, head at outlet, full dilation, ruptured membranes, adequate pelvis, known estimated fetal weight, reassuring or correctable tracing, effective contractions, empty bladder, good analgesia, pediatrician present, and a senior operator ready. Here operative vaginal delivery may be the most elegant answer: it avoids the morbidity of pushing a deeply engaged head back up through an abdominal incision and can complete birth safely.

The difference between those two cases is not enthusiasm. It is anatomy, mechanics, skill, and contingency.

Documentation is part of that discipline. It is not merely medicolegal self-protection. It is clinical reasoning made visible. The Israeli position paper asks documentation of the reason for instrumental delivery, explanation and consent, appropriate indications and absence of contraindications, fetal head position and direction, fetal monitoring description, estimated weight, instrument used, number of pulls, duration, detachments, episiotomy if performed, lacerations, and staff present. The local protocol similarly expects documentation of the advantages and disadvantages, explanation, consent, and procedural details.

Good documentation forces the clinician to answer the questions that should have been answered before traction began.

Why is delivery needed?

Why is vaginal delivery not prevented?

What is the bony station, not the caput?

What is the position?

Is the cervix fully dilated?

Are membranes ruptured?

Is the pelvis adequate for the estimated fetal weight?

Is analgesia adequate?

Is the bladder empty?

Who is the operator?

Where is the senior physician?

Is the pediatrician present?

Is cesarean readiness real?

What will count as failure?

Those questions are the procedure.

Let us finish by returning to the larger course.

The first modules taught diagnostic patience: do not call failed induction too early, do not call active-phase arrest before the six-centimeter gate, do not treat the second stage as a timer without mechanics. Module 5 added the fetal constraint: patience is allowed only while oxygenation and reserve permit it.

Module 6 adds the technical constraint: when patience ends, intervention must match anatomy.

Operative vaginal delivery is not a shortcut. It is not a consolation prize before cesarean. It is not a test pull to see what happens. It is a skilled procedure for a selected patient, with known station, known position, adequate pelvis, appropriate estimated fetal weight, full dilation, ruptured membranes, consent, analgesia, empty bladder, neonatal support, cesarean backup, and an operator prepared to abandon at the first evidence that the mechanics are wrong.

Second-stage cesarean is also not failure. It is the correct operation when vaginal delivery is unsafe, unlikely, contraindicated, or failed. It carries its own morbidity, especially with a deeply impacted head, but sometimes it is exactly the procedure that protects mother and fetus.

So the final synthesis is this.

Before applying an instrument, say the decision out loud: the indication, the station, the position, the pelvis, the estimated fetal weight, the fetal status, the operator, the backup plan, and the stopping rule. If you cannot say those things, you are not ready to pull. If you can say them, and the anatomy agrees, operative vaginal delivery remains one of the most powerful and beautiful forms of obstetric judgment: intervention that is decisive precisely because it is constrained.